Template Week 5 – Operating Systems

Student number: 585361

Assignment 5.1: Unix-like

a) Find out what the difference is between UNIX and unix-like operating systems?

UNIX is een merknaam die niet zomaar gebruikt mag worden. Operating systems die UNIX gebruiken moeten aan strenge voorwaarden voldoen om de naam UNIX te mogen gebruiken. Unix-like systems gebruiken de basis van UNIX maar hoeven verder niet te voldoen aan de strenge eisen van UNIX. Unix-like heeft hierdoor dus ook niet de licentie om UNIX genoemd te worden.

b) Study the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources often contain more detailed information about these individuals.

Ken Thompson: Ken is één van de 2 bedenkers en ontwerpers van UNIX, één van de bedenkers van de programmeertaal b en bijvoorbeeld de schaakmachine Belle. Thompson heeft bijna zijn hele leven gewerkt voor het bedrijf Bell Labs. Hij maakt verschillende kleine tools die uiteindelijk leiden tot het ontstaan van UNIX. Thompson maakte deze tools vanaf scratch op een minicomputer die hij had gevonden. De programmeertaal b is de taal die speciaal voor UNIX is ontwikkeld door Thompson. Veel van de hedendaagse OS worden gebasseerd op UNIX waardoor je kan zeggen dan Thompson één van de grondleggers is voor onze hedendaagse computers.

Dennis Ritchie: Ritchie is de levenslange collega van Thompson. Samen waren zij goed voor de ontwikkeling van UNIX. Ritchie haalde twee diplomma's aan Harvard, eentje in physics en eentje in applied mathematics. Ritchie zorgde ervoor dat de programmeertaal b (Thompson) werd vernieuwd naar de taal c. Door Ritchie kon UNIX worden gebruikt op andere platformen en PC's. zelfs vandaag de dag wordt c nog gebruikt als applicatie, OS en embedded system. C heeft ook veel andere programmeer talen van het heden beïnvloed.

Bill Joy: Joy is één van de oprichters van SUN, een bedrijf dat computers en computersystemen ontwikkelde en verkocht. In zijn studententijd was hij onderdeel van het begin van BSD UNIX en de tekst editor vi. Joy werkte dus aan zijn eigen UNIX system BSD. Bij SUN was Joy nog een inspiratie op de taal JAVA.

Richard Stallman: Stallman ontwikkelde GNU, een volledig vrij besturingssysteem. Dit deed hij omdat hij vond dat er onrechtvaardige dingen gebeurde MIT op zijn school. Stallman heeft geen huis, auto, vrouw of kinderen zodat hij alles kan geven aan GNU, het is voor hem echt een morele kwestie geworden. Hij geeft alle code vrij zodat iedereen kan werken aan zijn project. Dat mag allemaal zolang je de aanpassingen maar openbaar maakt. Stallman heeft dus veel invloed gehad op de vrije software ontwikkeling.

Linus Torvalds: Torvalds staat bekend voor het ontwikkelen van LinuxKernel. Deze code verspreid hij onder de GPL. Zijn kernel was de toevoeging aan GNU-project waar nog een kernel ontbrak. Ook Torvalds wilt graag helpen aan de ontwikkeling van software.

c) What is the philosophy of the GNU movement?
GNU gaat over de 'vrijheid' van software. Volgens Stallman moet elke gebruike de volgende vier dingen kunnen met software: gebruiken voor elk doel, besturen en aanpassen, verspreiden en verbeteren. Iedereen moet kunnen helpen ontwikkelen van de software en het vrij uit kunnen gebruiken.

d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement? Please explain your answer.
Ja, Ubuntu is ook open source. De code is voor iedereen te bekijken, gebruiken en aan te passen. Het is erg veilig en voldoet precies aan de eisen van de GNU movement.

e) Find out what is the Windows Subsystem for Linux?
WSL, wordt gebruikt zodat Linux kan worden gebruikt zonder externe VM.

f) Find out, which operating system family belongs to Android, iOS and ChromeOS?

Android = unix-like (linus-kernel)

iOS = UNIX (BSD0

ChromeOs = unix-like (linux-kernel)

Assignment 5.2: Supercomputers and gameconsoles

a) Research on this site what supercomputers are used for and write a short summary of it:
https://www.computerhistory.org/timeline/search/?q=Supercomputer
Supercomputers worden voornamelijk gebruikt in vak gebieden waar zware berekeningen gedaan worden.

b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for?

De playstation 3 cluster is een groep van ps3 die samenwerken als één groot computersysteem. Met de playstations kunnen krachtige berekeningen worden gemaakt die normaal veel duurder zijn. Elke playstation voert één kleine berekening uit die vervolgens samen worden gevoegd.

c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example:
https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster
What specific operating system is running on this cluster?
Oracle Linux for ARM

d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list.
https://www.top500.org/lists/top500/list/2023/06/

Ik denk het niet, de langzaamste van de snele supercomputers hebben ongeveer 45.000 cores. Elke raspberry pi heeft er vier dus dan kom je op 4 X 1050 is 4200. Dat is nog een aanzienlijk stuk minder.

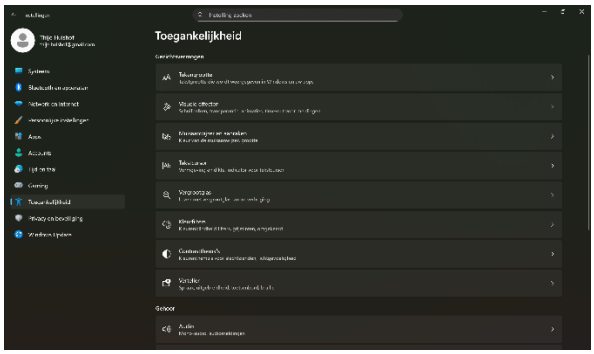
e) What CPU architecture is used for the PlayStation 5 and Xbox Series X?
What operating systems run on these consoles?
What conclusion can you draw from the answer to the previous question?

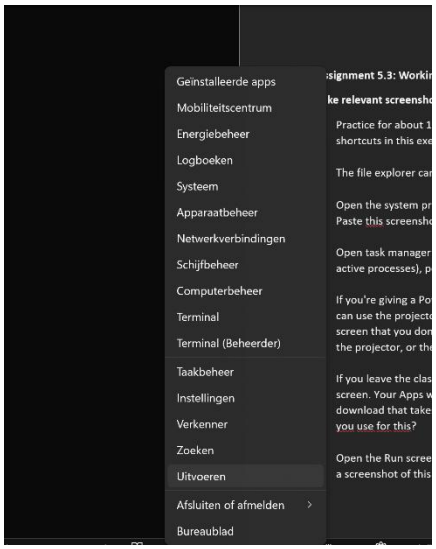
AMD zen 2 CPU architecture
De ps5 gebruikt de freeBSD en de xbox een variant van Microsoft windows.
Dat verschillende OS samen kunnen met één CPU.

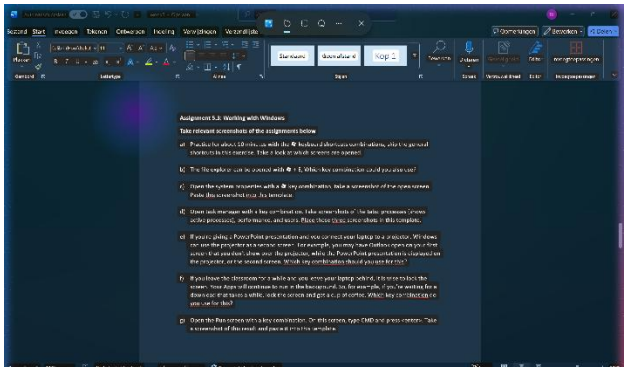
Assignment 5.3: Working with Windows

Take relevant screenshots of the assignments below

a) Practice for about 10 minutes with the ⊞ keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.



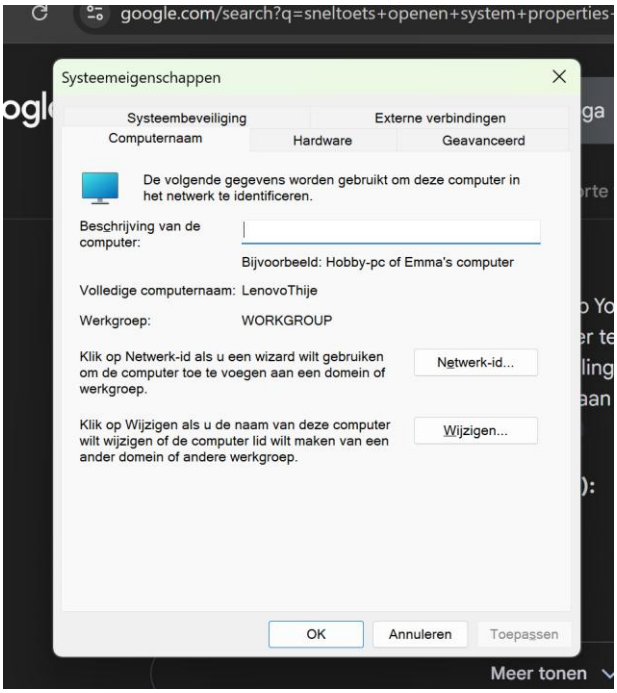




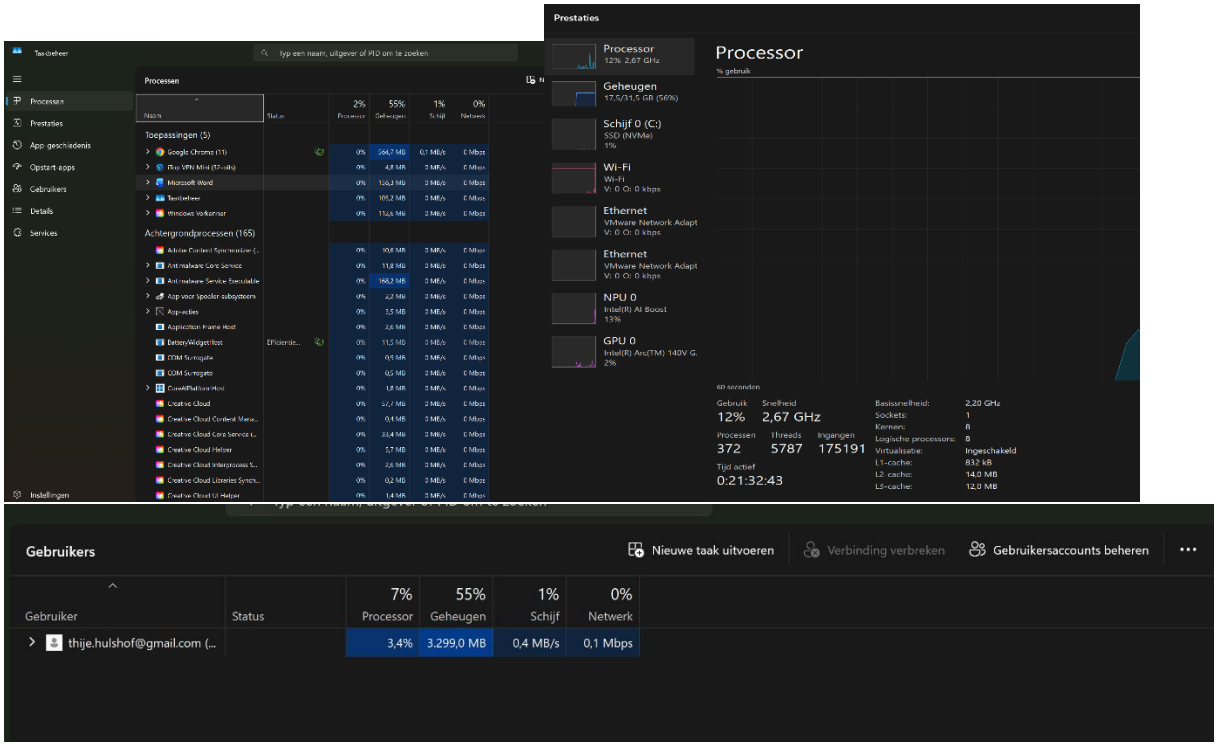
1 = windows + i
2 = windows + x
3 = windows + w
4 = windows + c, opent co-pilot
5 = windows + l, laptop vergrendeld

b) The file explorer can be opened with ⊞ + E, Which key combination could you also use?
Control O

c) Open the system properties with a ⊞ key combination, take a screenshot of the open screen. Paste this screenshot into this template.



d) Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.

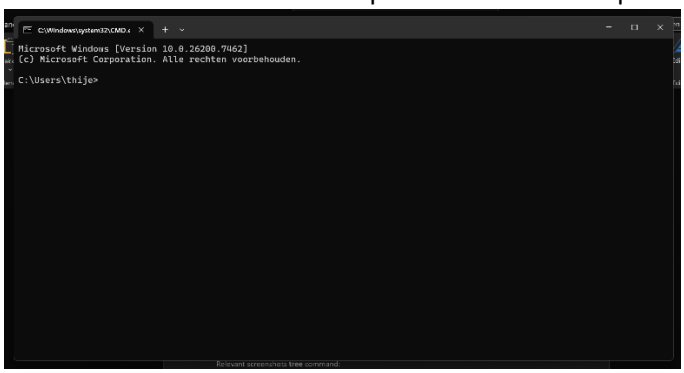


e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?
Windows + p

f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?

Windows + l

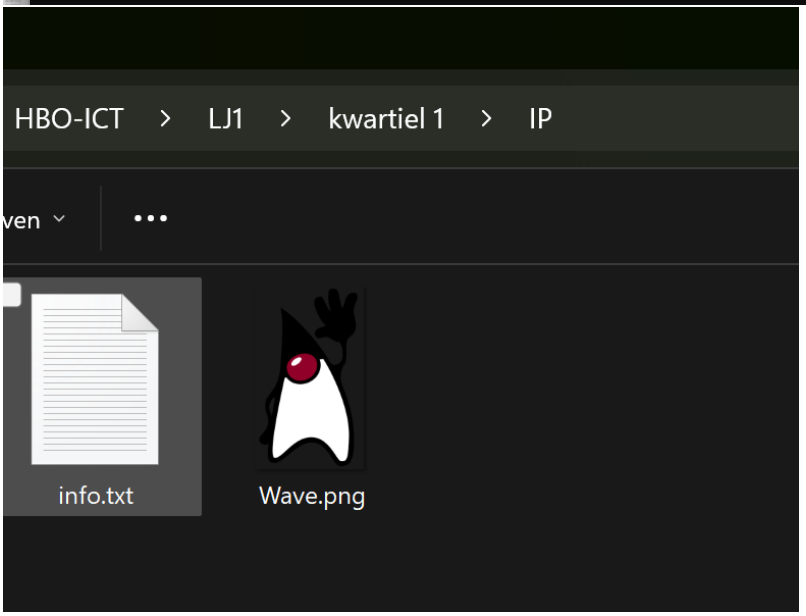
g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.



Working in the File Explorer

Relevant screenshots **copy** command:

```
C:\Saxion> copy Wave.png "HBO-ICT\LJ1\kwartiel 1\IP"
        1 file(s) copied.

C:\Saxion>copy Plug.png "HBO-ICT\LJ1\kwartiel 1\fundamentals"
        1 file(s) copied.

C:\Saxion>copy Tumble.png "HBO-ICT\LJ1\kwartiel 1\organisatie en it"
        1 file(s) copied.
```

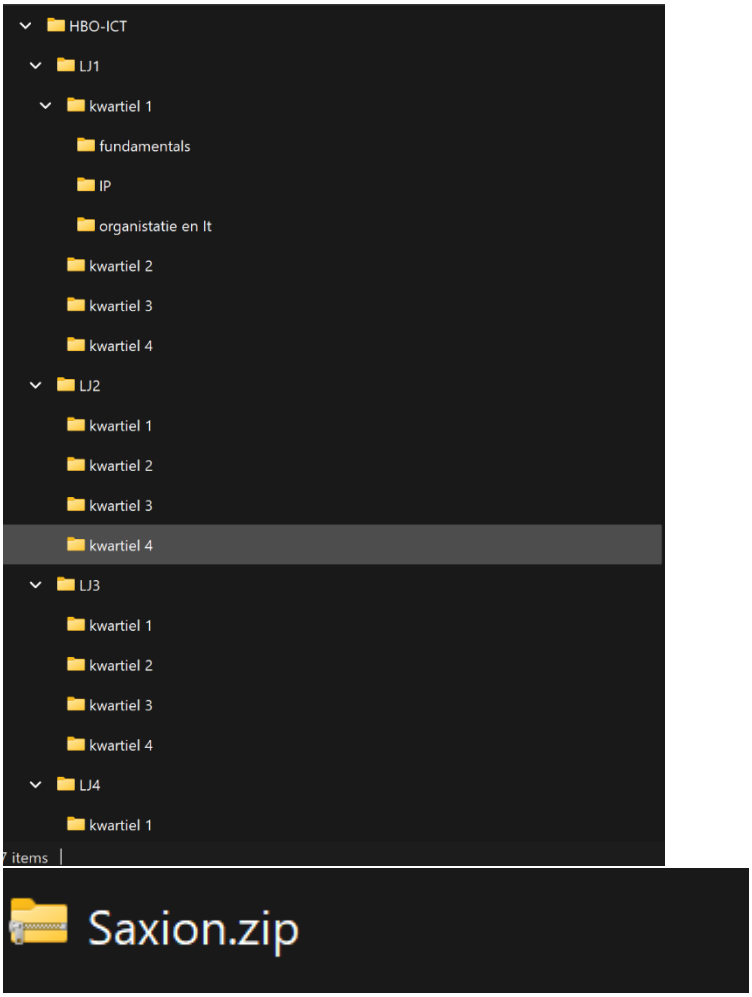


Voorbeeld van IP

Relevant screenshots **tree** command:

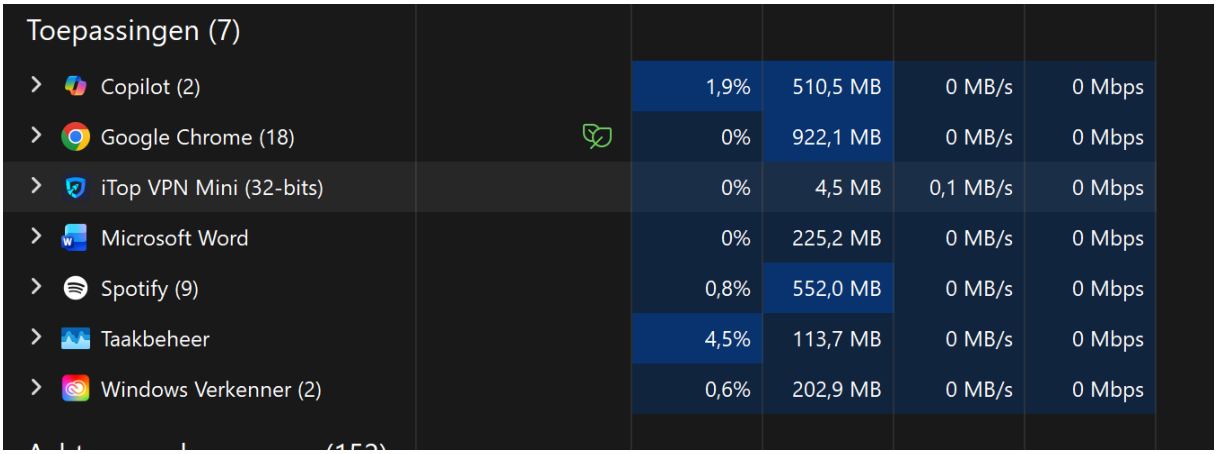
```
Folder PATH listing for volume Windows-SSD
Volume serial number is 505D-6667
C:.
└───HBO-ICT
    ├───LJ1
    │   ├───kwartiel 1
    │   │   ├───fundamentals
    │   │   ├───IP
    │   │   └───organistatie en It
    │   ├───kwartiel 2
    │   ├───kwartiel 3
    │   └───kwartiel 4
    ├───LJ2
    │   ├───kwartiel 1
    │   ├───kwartiel 2
    │   ├───kwartiel 3
    │   └───kwartiel 4
    ├───LJ3
    │   ├───kwartiel 1
    │   ├───kwartiel 2
    │   ├───kwartiel 3
    │   └───kwartiel 4
    └───LJ4
        ├───kwartiel 1
        ├───kwartiel 2
        ├───kwartiel 3
        └───kwartiel 4

C:\Saxion> echo %username%
thije
```

Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



Terminating Processes

Relevant Screenshots Task Manager Window:

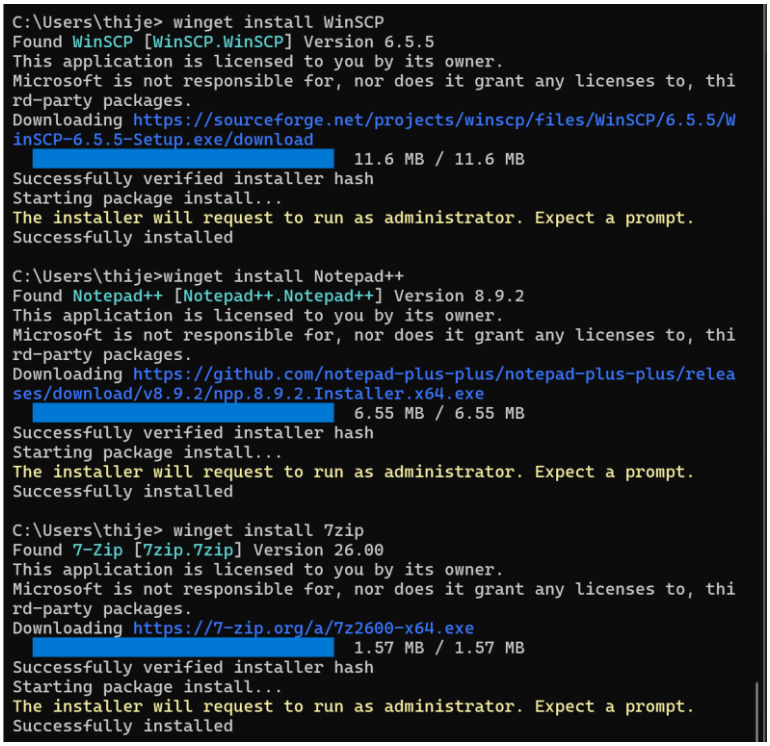


Toepassingen (7)				
Copilot (2)	1,9%	510,5 MB	0 MB/s	0 Mbps
Google Chrome (18)	0%	922,1 MB	0 MB/s	0 Mbps
iTop VPN Mini (32-bits)	0%	4,5 MB	0,1 MB/s	0 Mbps
Microsoft Word	0%	225,2 MB	0 MB/s	0 Mbps
Spotify (9)	0,8%	552,0 MB	0 MB/s	0 Mbps
Taakbeheer	4,5%	113,7 MB	0 MB/s	0 Mbps
Windows Verkenner (2)	0,6%	202,9 MB	0 MB/s	0 Mbps

Install Software

Relevant screenshots that the following software is installed with winget:

- WinSCP
- Notepad++
- 7zip



```
C:\Users\thije> winget install WinSCP
Found WinSCP [WinSCP.WinSCP] Version 6.5.5
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, thi
rd-party packages.
Downloading https://sourceforge.net/projects/winscp/files/WinSCP/6.5.5/W
inSCP-6.5.5-Setup.exe/download
  ██████████████████████████████  11.6 MB / 11.6 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator. Expect a prompt.
Successfully installed

C:\Users\thije>winget install Notepad++
Found Notepad++ [Notepad++.Notepad++] Version 8.9.2
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, thi
rd-party packages.
Downloading https://github.com/notepad-plus-plus/notepad-plus-plus/relea
ses/download/v8.9.2/npp.8.9.2.Installer.x64.exe
  ██████████████████████████████  6.55 MB / 6.55 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator. Expect a prompt.
Successfully installed

C:\Users\thije> winget install 7zip
Found 7-Zip [7zip.7zip] Version 26.00
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, thi
rd-party packages.
Downloading https://7-zip.org/a/7z2600-x64.exe
  ██████████████████████████████  1.57 MB / 1.57 MB
Successfully verified installer hash
Starting package install...
The installer will request to run as administrator. Expect a prompt.
Successfully installed
```

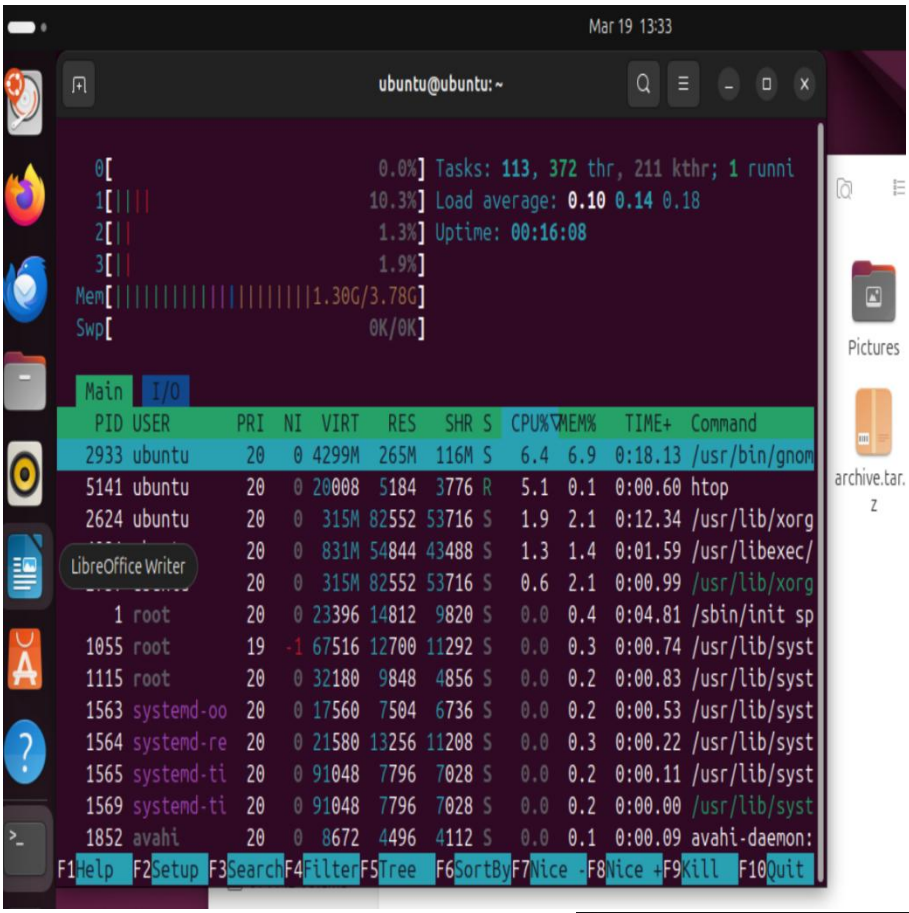
Assignment 5.4: Working with Linux

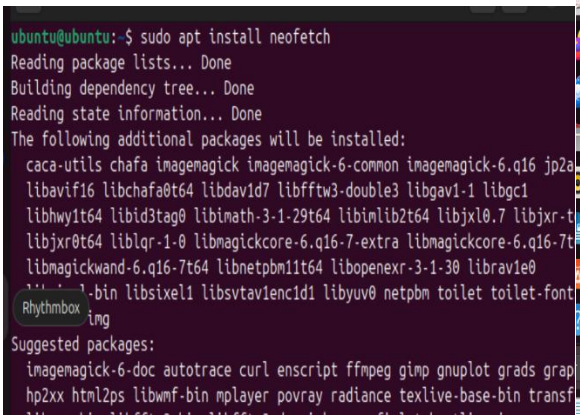
```
ubuntu@ubuntu:~$ nano test.txt
ubuntu@ubuntu:~$ cp ~/test.txt ~/Documents/
ubuntu@ubuntu:~$ cd ~/Documents
ubuntu@ubuntu:~/Documents$ ls
test.txt
ubuntu@ubuntu:~/Documents$
```

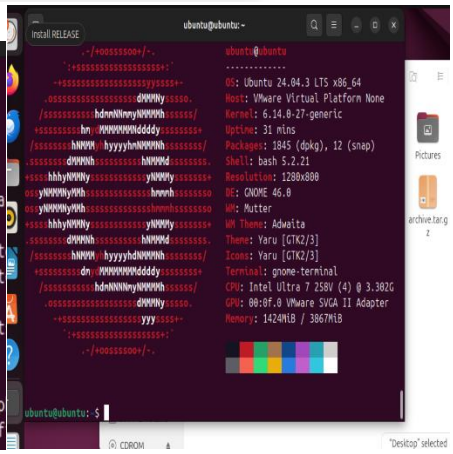
```
ubuntu@ub
ubuntu@ubuntu:~$ cd /etc
ubuntu@ubuntu:/etc$ cd ~
ubuntu@ubuntu:~$
```

```
Firefox ubuntu:~$ tar -cvf archive.tar test.txt
test.txt
ubuntu@ubuntu:~$ tar -xvf archive.tar
test.txt
ubuntu@ubuntu:~$ tar -cvzf archive.tar.gz text.txt
```

```
ubuntu@ubuntu:~$ sudo apt install htop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  lm-sensors
The following NEW packages will be installed:
  htop
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 171 kB of archives.
After this operation, 434 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 htop amd64 3.3.0-4build1
 [171 kB]
Fetched 171 kB in 0s (1,002 kB/s)
```







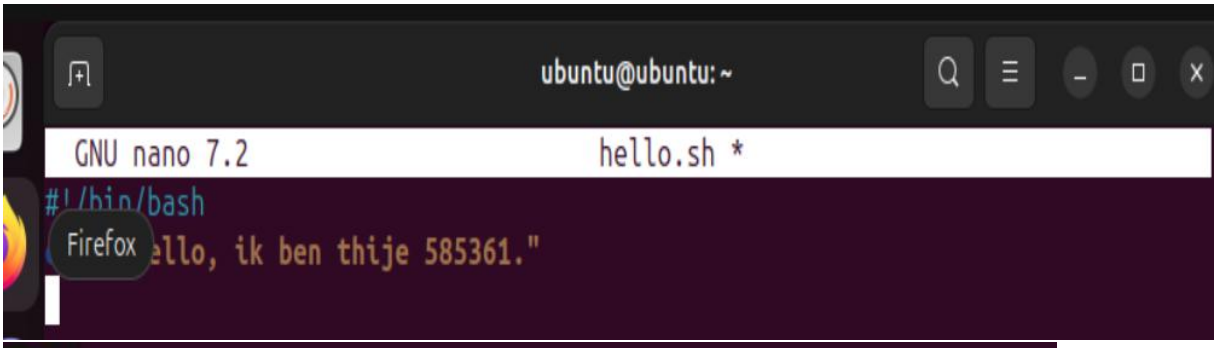
Linux heft een home directory terwijl windows meerdere schijven heeft (aangegeven met letters).

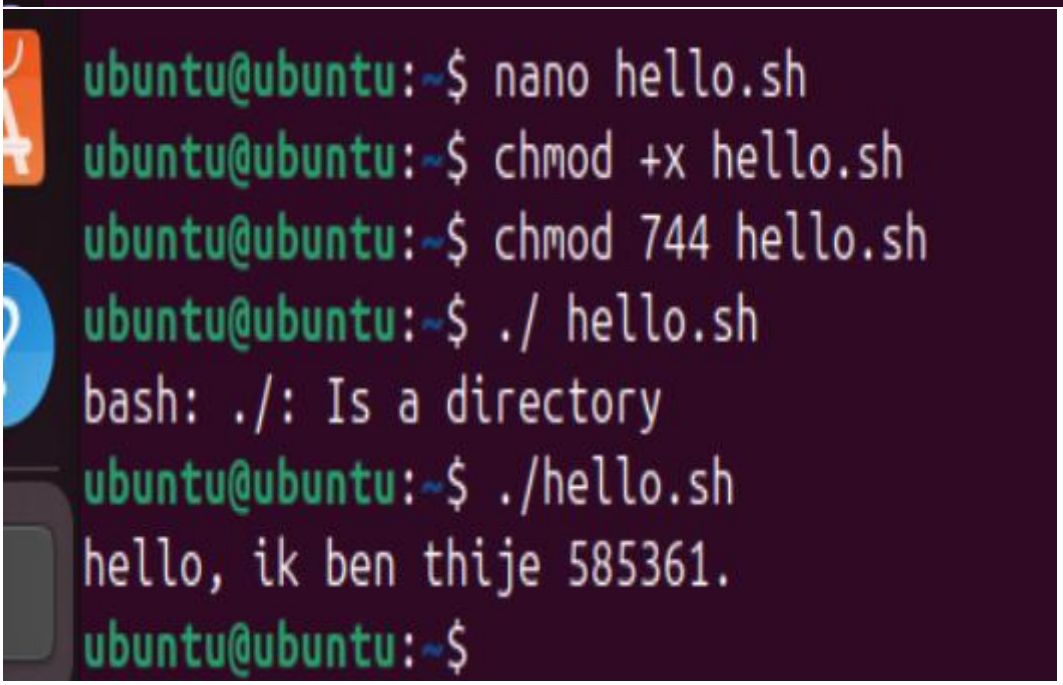
/etc bevat configuratiebestanden

Je ziet in htop de prestaties van je apparaat.

Assignment 5.5: Users and permissions on Linux

Relevant screenshots + motivation





744 staat voor de bepaalde users die het bestand kunnen gebruiken

Assignment 5.6: View the contents of files

Relevant screenshots + motivation

Cat toont regels

Wc telt regels, woorden, karakters

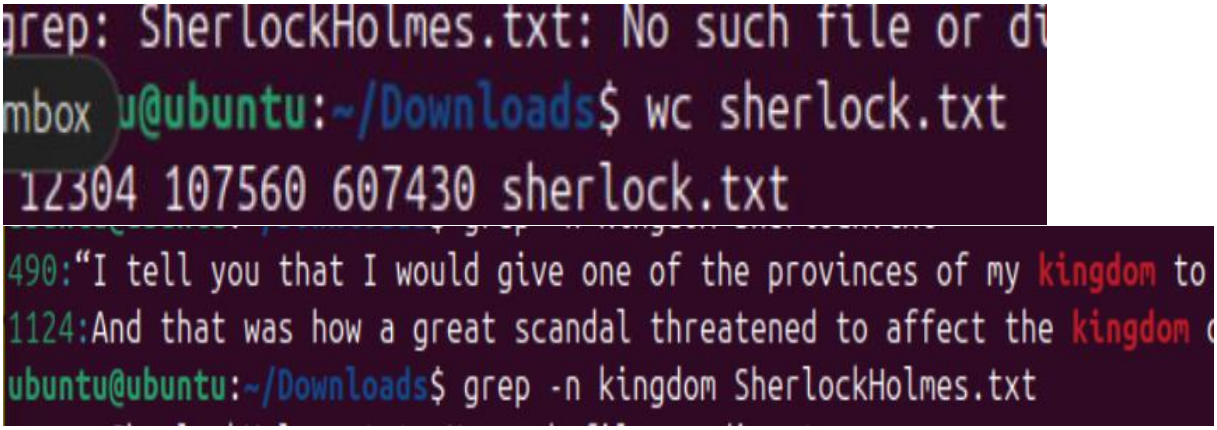
Less = scrollbare viewer

Head = eerste regel

Tail = laatste regel

Grep = zoeken in tekst

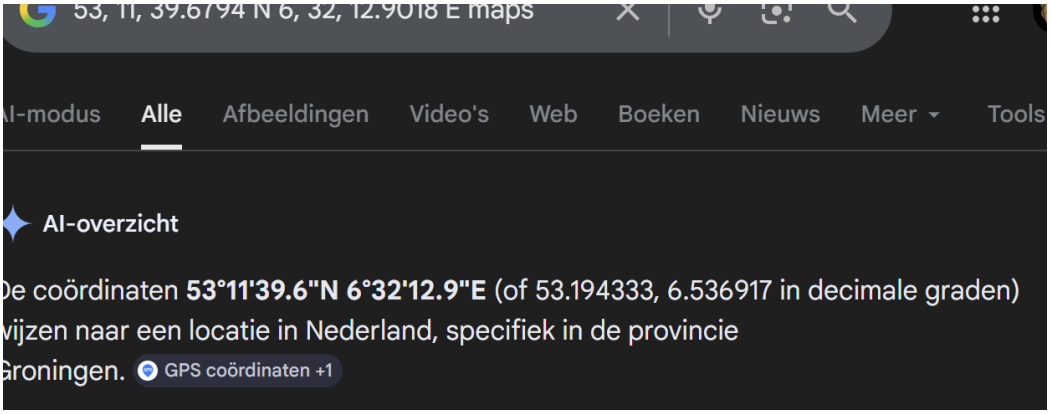
Bestand heeft 12304 regels, 107560 woorden en 607430 tekens.



Assignment 5.7: Digital forensics

Relevant screenshots + motivation

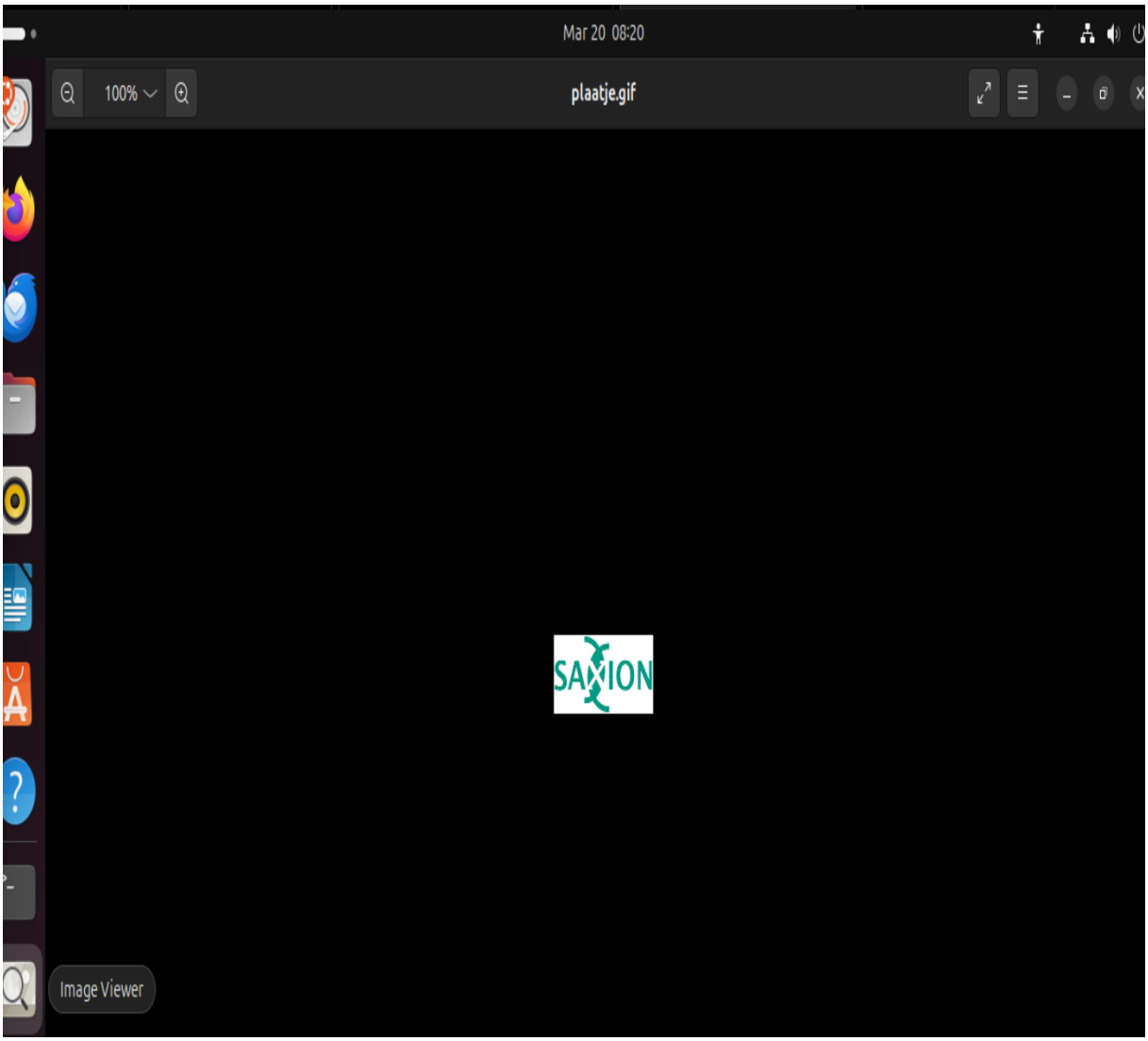
Motorola g6 play



```
ubuntu@ubuntu:~/Downloads$ exif oldcar.jpeg
EXIF tags in 'oldcar.jpeg' ('Motorola' byte order):
-------------------+-------------------------------------------------------
Tag                |Value
-------------------+-------------------------------------------------------
Manufacturer       |motorola
Model              |moto g(6) play
X-Resolution       |72
Y-Resolution       |72
Resolution Unit    |Inch
Software           |aljeter-user 9 PPPS29.55-35-18-7 6a0d0 release-keys
Date and Time      |2020:11:07 15:08:57
YCbCr Positioning  |Centered
Compression        |JPEG compression
X-Resolution       |72
Y-Resolution       |72
Resolution Unit    |Inch
Exposure Time      |1/33 sec.
F-Number           |f/2.0
Exposure Program   |Normal program
ISO Speed Ratings  |64
Exif Version       |Exif Version 2.2
Date and Time (Origi|2020:11:07 15:08:57
Date and Time (Digit|2020:11:07 15:08:57
```

```
ubuntu@ubuntu:~/Downloads$ mv oldcar.jpeg oldcar
ubuntu@ubuntu:~/Downloads$ file oldcar
oldcar: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment
length 16, Exif Standard: [TIFF image data, big-endian, direntries=10, manufactu
rer=motorola, model=moto g(6) play, xresolution=160, yresolution=168, resolution
unit=2, software=aljeter-user 9 PPPS29.55-35-18-7 6a0d0 release-keys, datetime=2
020:11:07 15:08:57, GPS-Data], baseline, precision 8, 4160x3120, components 3
ubuntu@ubuntu:~/Downloads$
```

Is nogsteeds een jpeg bestand




```
ubuntu@ubuntu:~/Downloads$ base64 -d email-base64.txt > plaatje.gif
```

Assignment 5.8: Steganography

Relevant screenshots + motivation

```
ubuntu@ubuntu:~/Downloads$ steghide extract -sf apple2.jpeg
Enter passphrase:
wrote extracted data to "message.txt".
ubuntu@ubuntu:~/Downloads$ cat message.txt
Hello class.
You have almost completed Week 5.
```

Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.
- Proof that you can restore the back-up image into an empty VM.

127.0.0.1

Ready? Save this file and export it as a pdf file with the name: **week5.pdf**